SRAI

EXECUTIVE BRIEF

VECTOR FOUNDATIONS

Why representation choices govern similarity, comparison and AI reasoning

Production Unit	PU-B01-C03
Book / Lesson	Book 1 / Lesson 3
Subject	Vector Foundations
Control status	**Approved controlled companion v1.0**

Scientific Reasoning and Artificial Intelligence (SRAI)
Created by Mbaye Kebe

Why this lesson matters

Vectors are the working language through which modern data systems represent observations, documents, images, model parameters, gradients and decision states. Once information becomes a vector, comparison and learning become geometric operations. That power is accompanied by a governance obligation: coordinate meaning, units, scaling, weighting and missing-data treatment determine the geometry and therefore influence the result.

The operational foundations

Foundation	Mathematical role	Operational implication
Representation	Ordered coordinates encode an object or state.	Define coordinate names, order, units, population and period.
Norms and distance	Quantify magnitude and separation.	Choose a metric whose meaning matches the decision; standardize incompatible scales.
Dot product and cosine	Measure alignment or directional similarity.	Treat high similarity as model-dependent evidence, not semantic identity or causation.
Projection	Separates aligned and orthogonal components.	State the chosen direction and verify the residual is orthogonal.
Span and basis	Define the directions available to a representation.	Check independence, rank and whether the basis excludes relevant structure.

Decision-system application

Consider a national monitoring dashboard whose state vector contains growth, inflation, food production, electricity access and health capacity. A distance or similarity score can compare countries or periods only after each coordinate has a stable definition. Inflation and deprivation may need their direction reversed for a common interpretation; large-unit indicators may dominate raw Euclidean distance; missing values and revisions may alter rankings; and policy weights embed value judgments. The resulting geometry is therefore a controlled analytical construction, not a neutral property of the countries themselves.

Controls leaders should require

- Approve a data dictionary fixing coordinate order, units, reference population and period.
- Record scaling, normalization and weighting parameters with an effective date and owner.
- Keep missing, uncertain and not-applicable states visible; do not silently convert them to zero.
- Test domain restrictions, including zero vectors and incompatible dimensions.
- Report sensitivity under plausible alternative transformations and weights.
- Attach the representation version and reasoned interpretation to every published similarity or ranking.
- Require human and domain validation before using embedding similarity or vector scores in consequential decisions.

Implications for AI and machine learning

Feature vectors, embeddings, attention queries and keys, hidden states and gradients all rely on vector operations. A technically correct model can still mislead when its representation is poorly defined or its geometry is interpreted beyond its domain. Vector governance therefore strengthens traceability, reproducibility and contestability at the point where raw information becomes computational structure.

What the controlled release provides

- An audited Chapter 3 publication covering vector representation, norms, similarity, projection, span, bases and scaling.
- A canonical executable notebook with environment reporting, numerical safeguards and invariant checks.
- Fifteen assessed exercises with complete solutions and safe Python reference implementations.
- A coordinated long-form YouTube lesson and controlled GitHub release package.
- Validation evidence, checksums and a release record linking every approved component.

Recommended executive use

Use this Production Unit as a design and governance checkpoint whenever an initiative ranks entities, compares profiles, retrieves similar documents, builds composite indicators, uses embeddings or applies distance-based learning. Before approval, ask the team to show the vector schema, transformations, domain checks, sensitivity results and the limits of interpretation.

Controlled resources

Lesson page	https://srai.mbayekebe.net/learn/pu-b01-c03/
YouTube lesson	https://www.youtube.com/watch?v=THvzO5_L-Zs
GitHub repository	https://github.com/mbayekebe/srai-book-01-mathematical-foundations
Canonical notebook	M1_N03_vector_foundations.ipynb
Production unit	PU-B01-C03

Sources and control basis

- PU-B01-C03 audited Chapter 3 controlled edition v1.0.
- M1_N03_vector_foundations.ipynb and tested srai_math vector utilities.
- PU-B01-C03 Chapter and Notebook Audit Report.
- NumPy, SciPy and scikit-learn primary documentation cited by the controlled chapter.